## Pre−Pandemic Data Only

antiGen
ESM 2 (35M)
BLOSUM62
EVEscape
ESM 2 (650M)

## antiGen Finetunes

antiGen
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (650M) partial FT
antiGen as Tranception full FT
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (650M) full FT
antiGen as EVE full FT

## Pandemic Data in Training Window

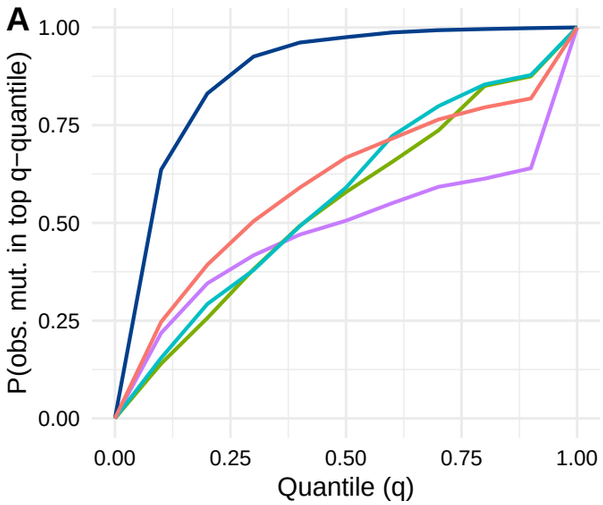
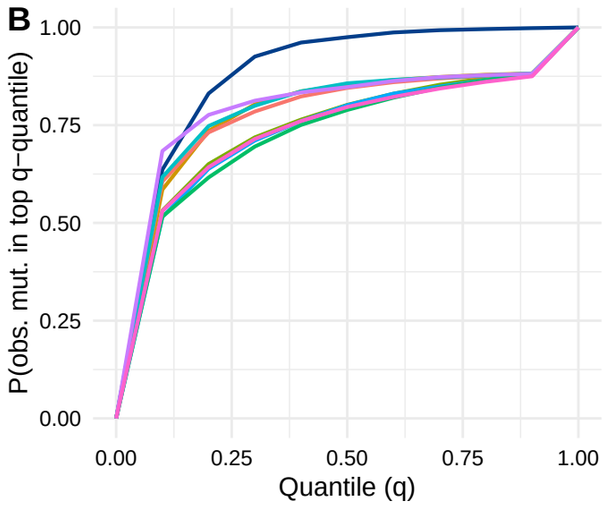
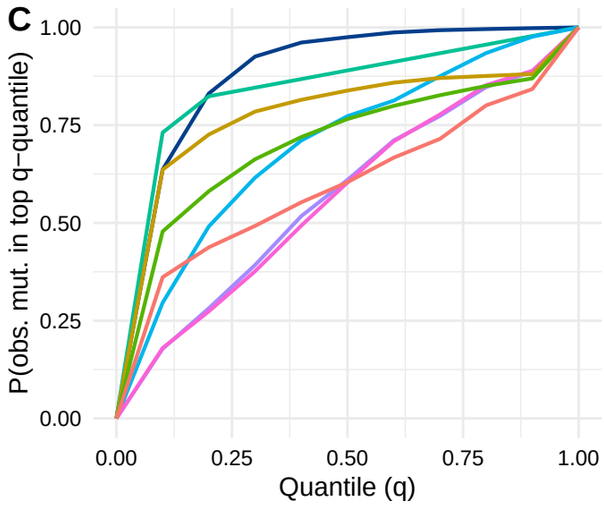
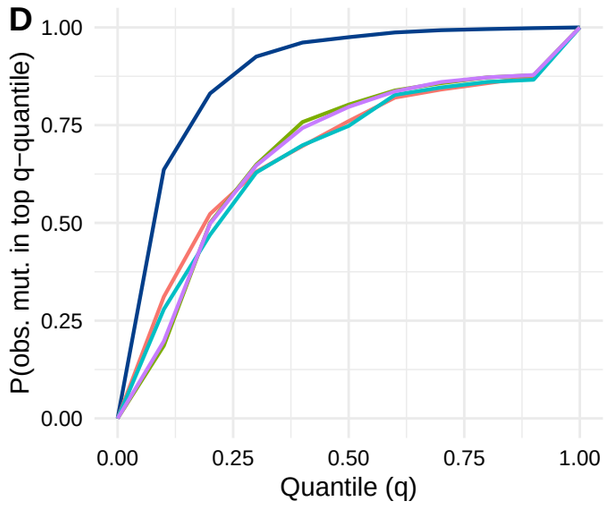
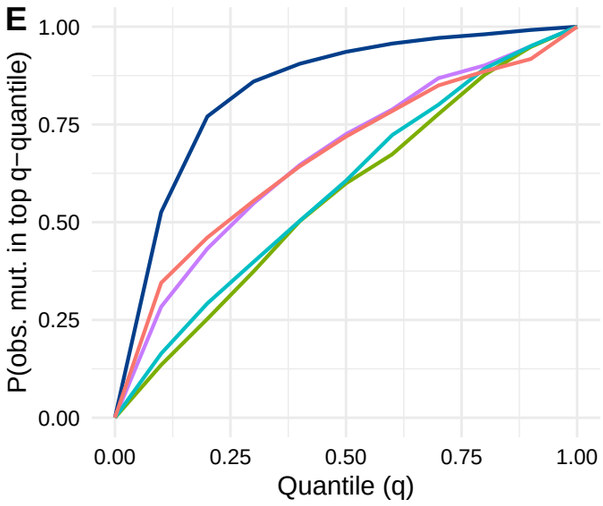
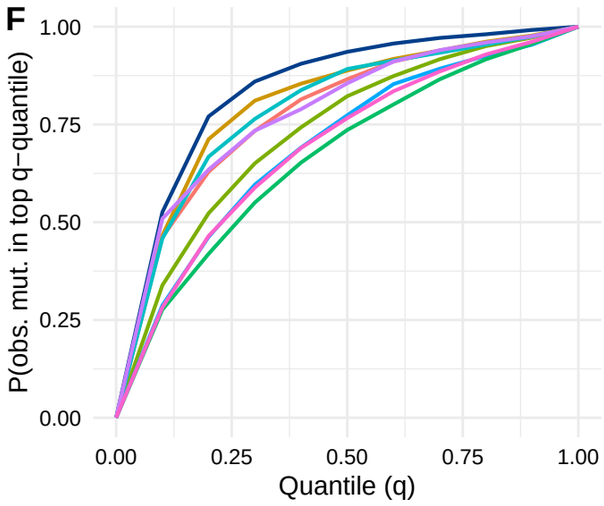
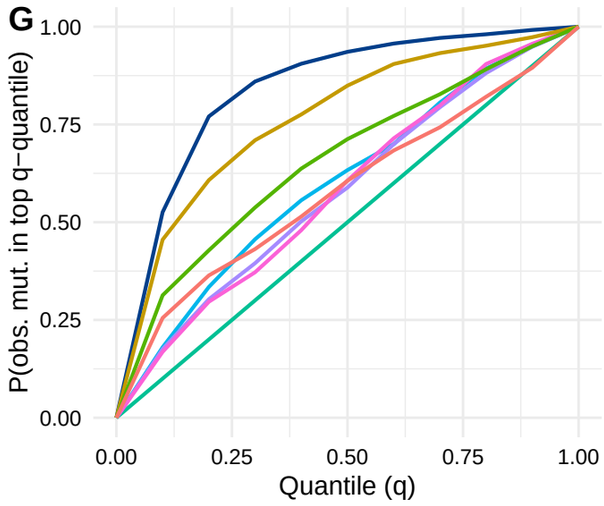
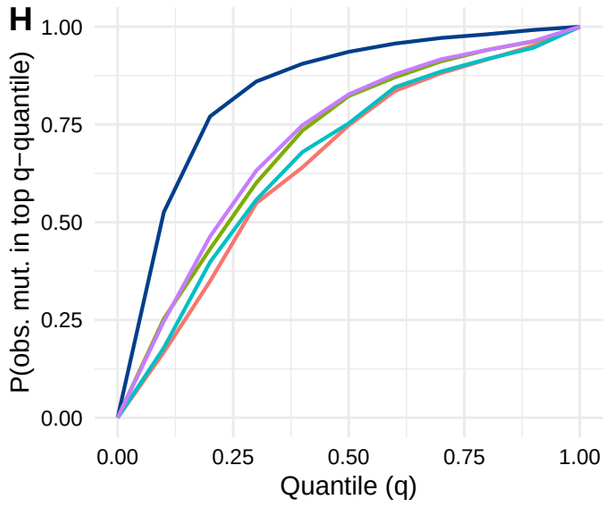
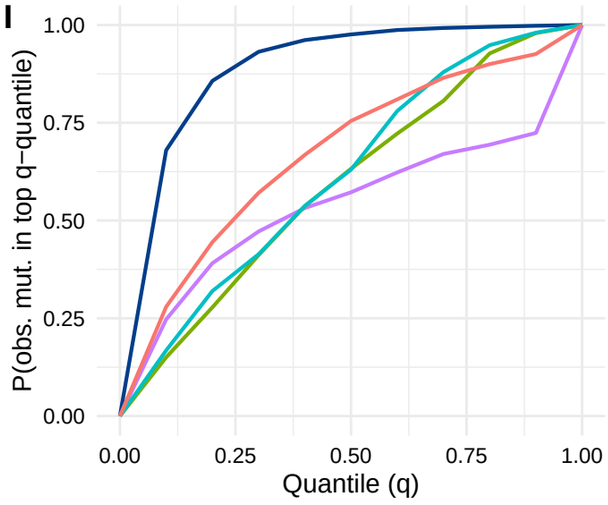
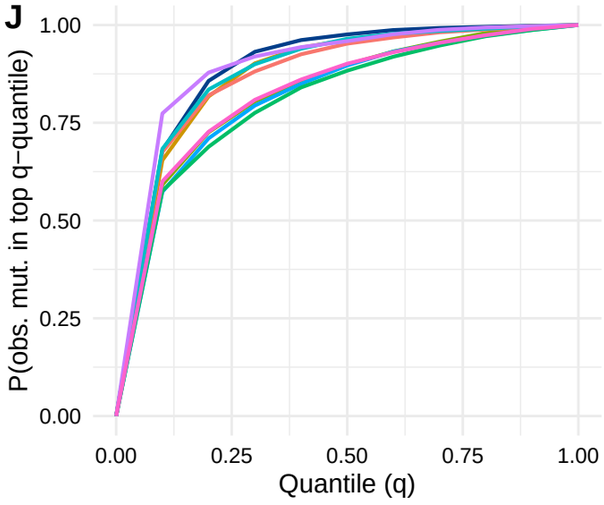
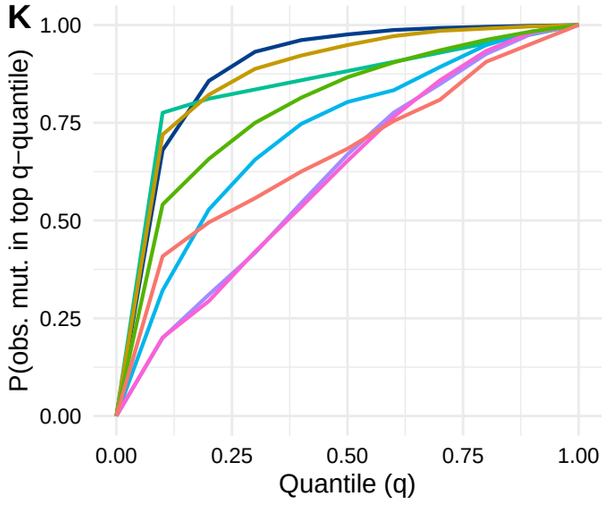
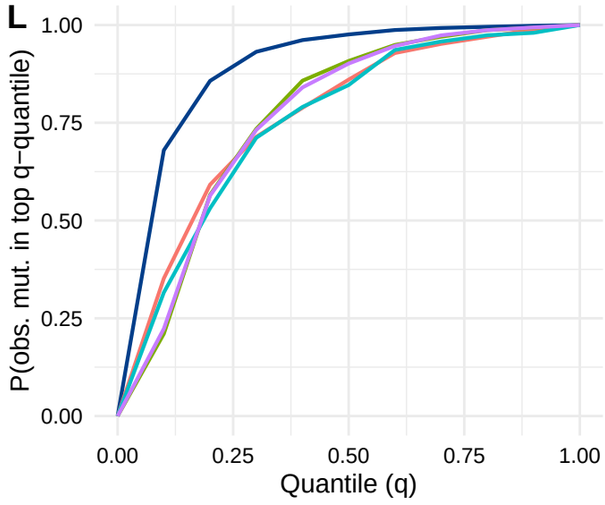
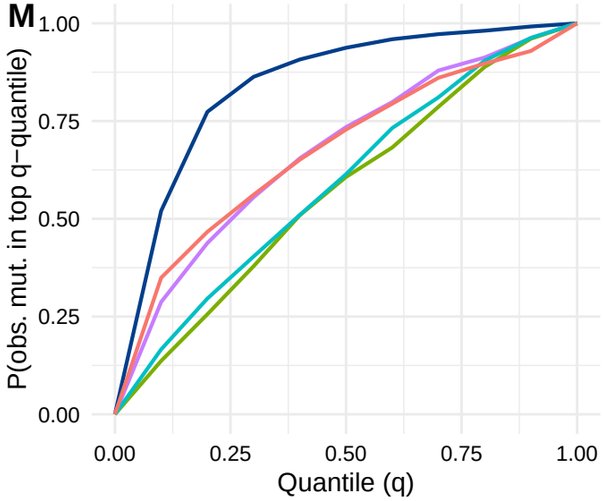
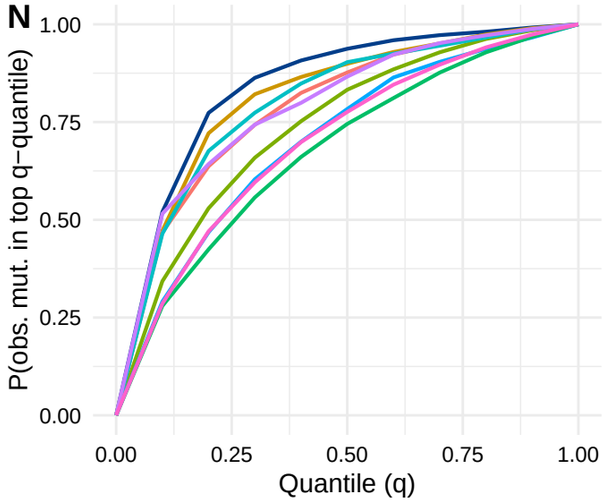
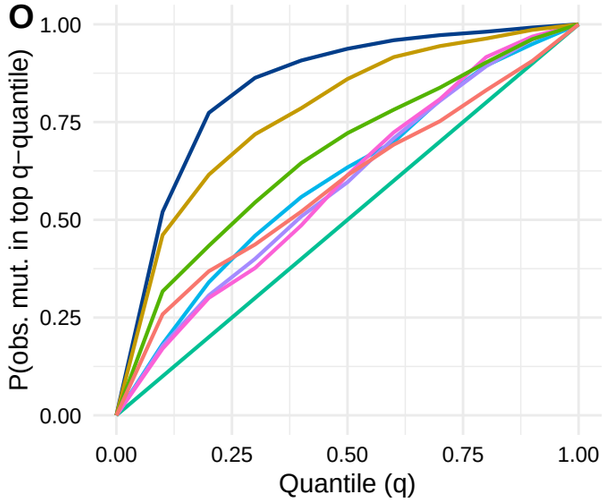
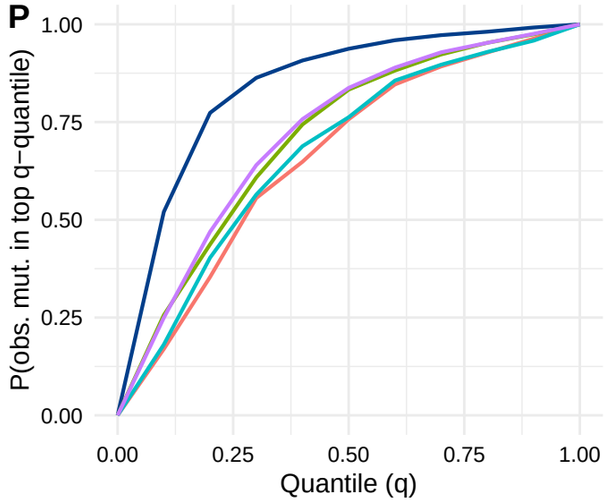
antiGen
PSSM Conservation
MSA Trans. (BA.2 eval)
Tranception
PSSM
MSA Transfomer
EVE
EVmutation

## Deep Mutational Scans

antiGen
BA.1 DMS (expression)
BA.2 DMS (expression)
BA.1 DMS (bind)
BA.2 DMS (bind)

**all mutations**

**de novo**

**gap/stop mask**

**de novo + gap/stop mask**